

SD/MMC MEMORY CARD FUNDAMENTALS

SD and MMC cards differ only in the bus topology and initialisation protocol

■ TULARAM MADGUJI BANSOD

With mobile devices taking off in a big way, sales of memory cards have also picked up. These tiny cards plug into your device and store digital data like photos, music, movies, documents and much more. You can get a 4GB MMC for Rs 300. But given the requirement for storing more and more digital data, you wonder why aren't they doing fast research to offer 32GB cards for Rs 300. However, I feel a day will come when 32GB SD cards will be available for Rs 300 or less—recall three years ago, a tiny card of 2 GB cost you as much as Rs 350.

You may ask what does SD/MMC stand for. SD means 'secure digital' and MMC means 'multimedia card.' You can insert these cards in your media player, PDA or digital camera. Their small size, relative simplicity, low power consumption and low cost make them an ideal solution for many applications.

Manufacturers

In August 1999, three big electronics firms—Matsushita Electric, SanDisk and Toshiba—decided to compete with Sony's MemoryStick. They focused on designing small memory cards that could be used by people to protect their digital information from accidental erasure. The card supported Digital Rights Management (DRM) and Secure Digital Music Initiative (SDMI) standards. SDMI is a forum of 200 IT companies that manufacture various multimedia



Fig. 1: Some SD/MMC cards



Fig. 2: Backsides of two SD/MMC cards

products.

In March 2001, the trio designed SD card specification. With this specification, SD/MMC cards became suitable for many different brands of cameras and media players.

The basics

SD/MMC cards have their own architecture and signals. These are universal low-cost, high-speed data storage cards. MMCs work at 20 MHz, while SD cards work at up to 25 MHz. The two memories work in two different modes: SD mode and serial peripheral interface (SPI).

You can have miniSD, microSD, read/write (RW), Flash, one-time programmable (OTP), multiple-time programmable (MTP) and read-only memory (ROM) cards. SD cards are available in speed ratings of 2 MB/s, 4 MB/s and 6 MB/s.

Fig. 3 shows the assignment of pin numbers on SD cards. The table shows signals and functions of pins in both modes. These signals are used for data communication and controls.

SD card signals in SD and SPI modes

The only difference between SD and MMC cards is in the bus topology and initialisation protocol. SD memory